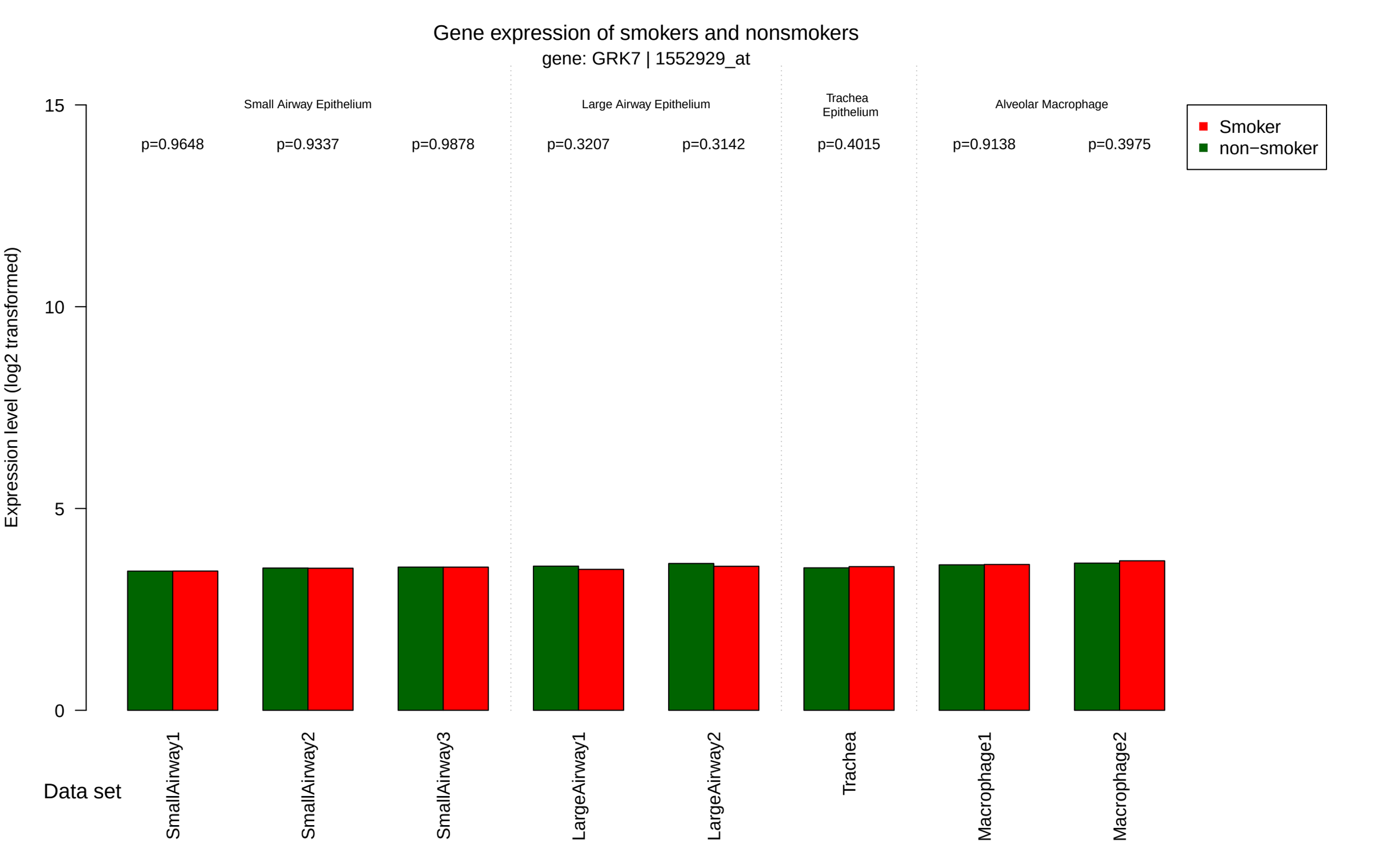
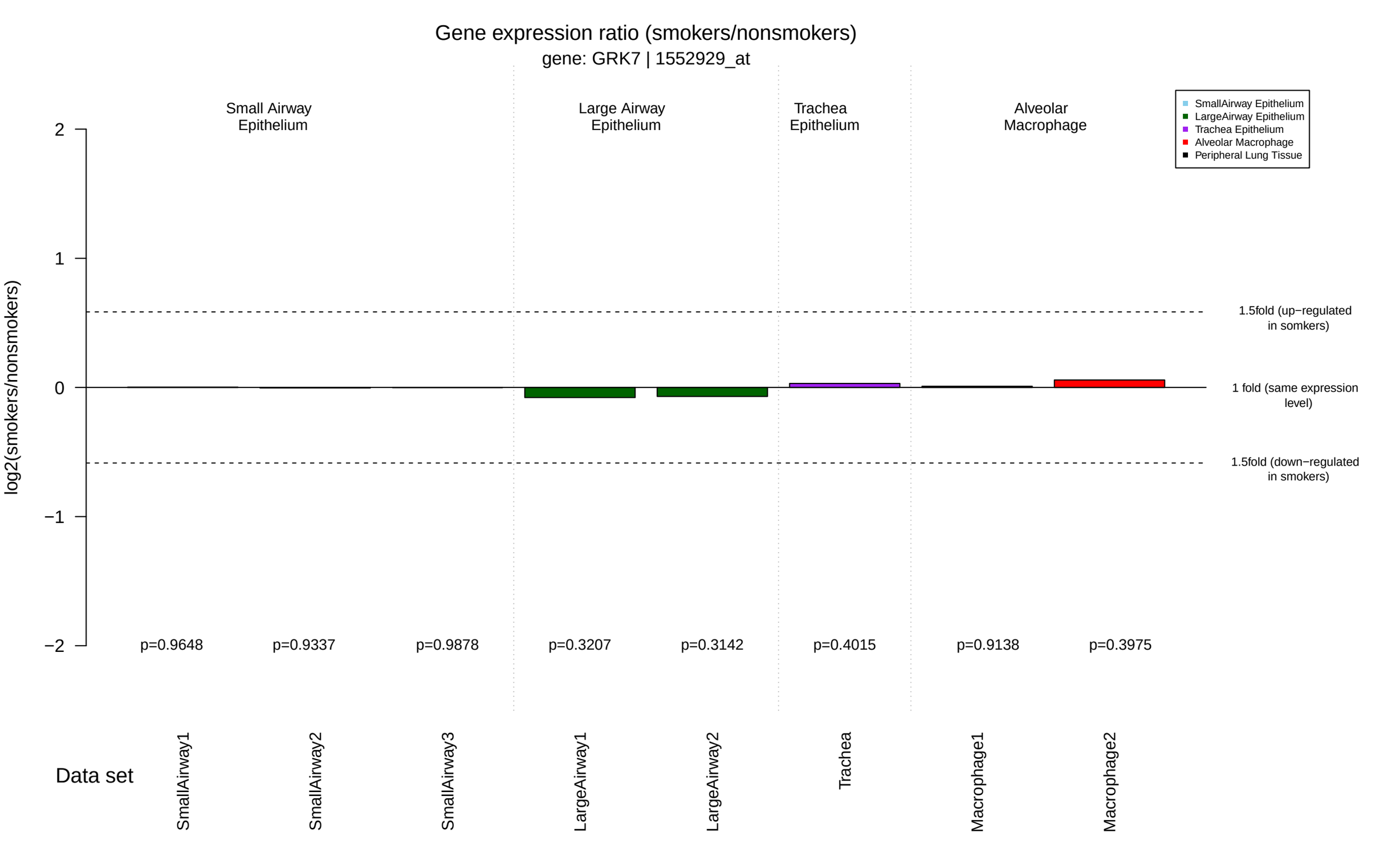
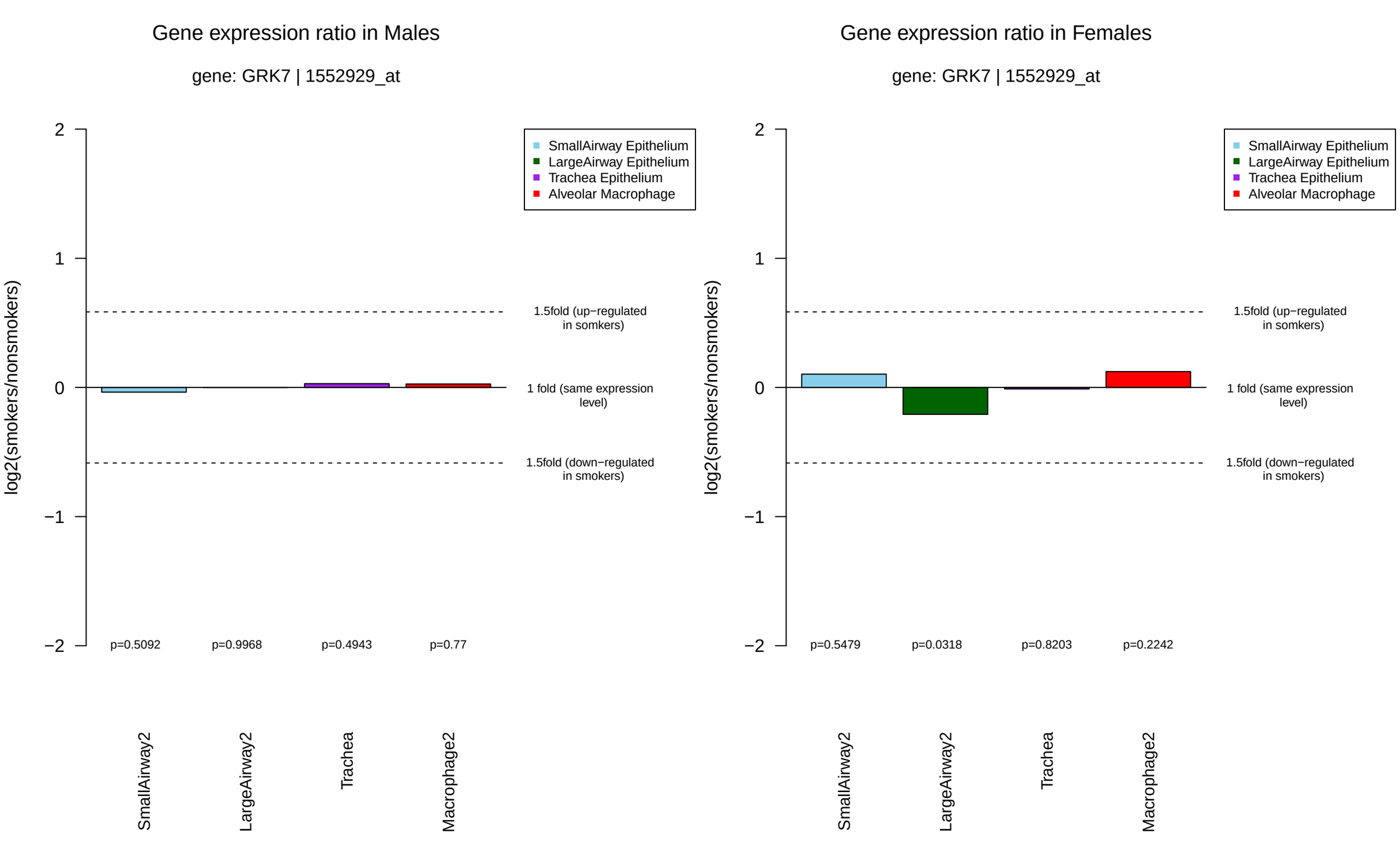
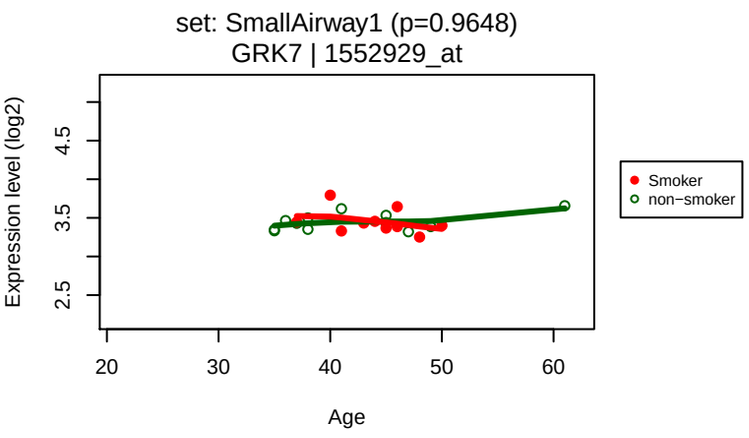
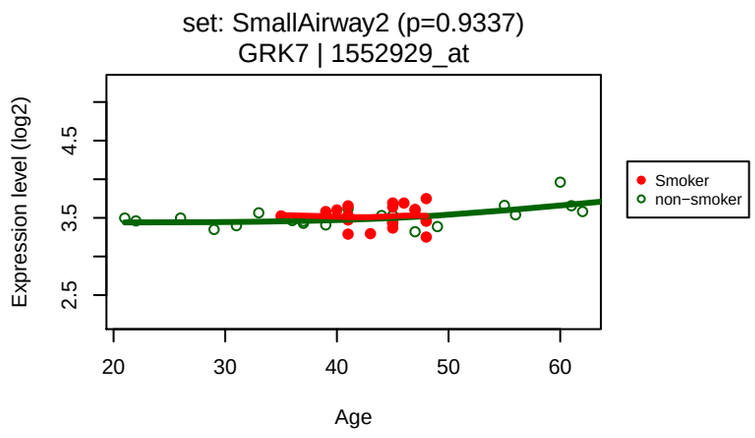
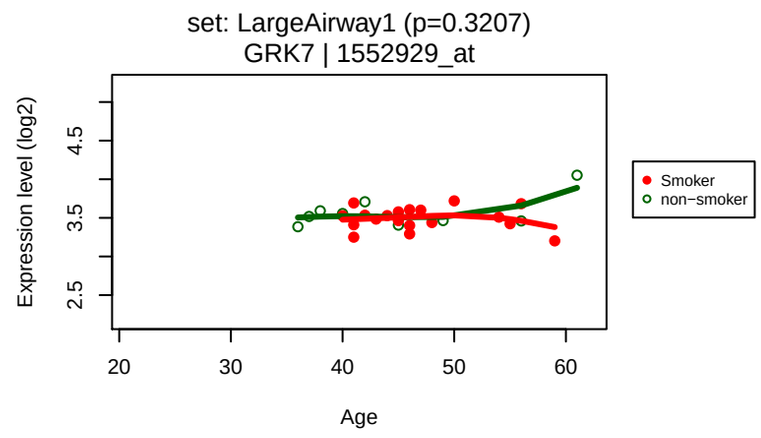

Gene expression of smokers and nonsmokers
gene: GRK7 | 1552929_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9648
p=0.9337
p=0.9878
p=0.3207
p=0.3142
p=0.4015
p=0.9138
p=0.3975
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

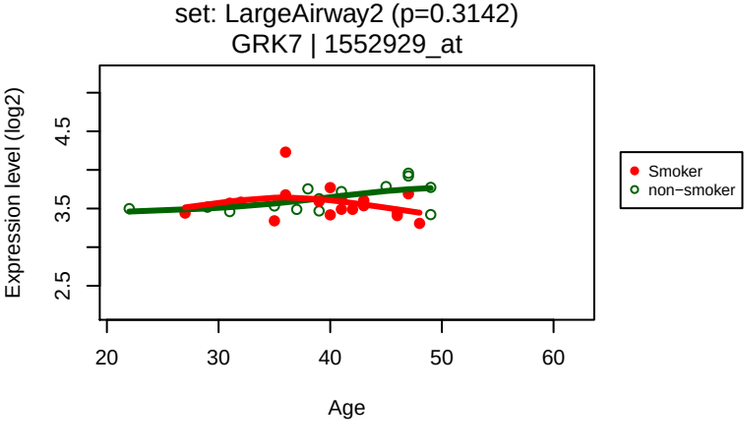

Gene expression ratio (smokers/nonsmokers)
gene: GRK7 | 1552929_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9648
p=0.9337
p=0.9878
p=0.3207
p=0.3142
p=0.4015
p=0.9138
p=0.3975
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

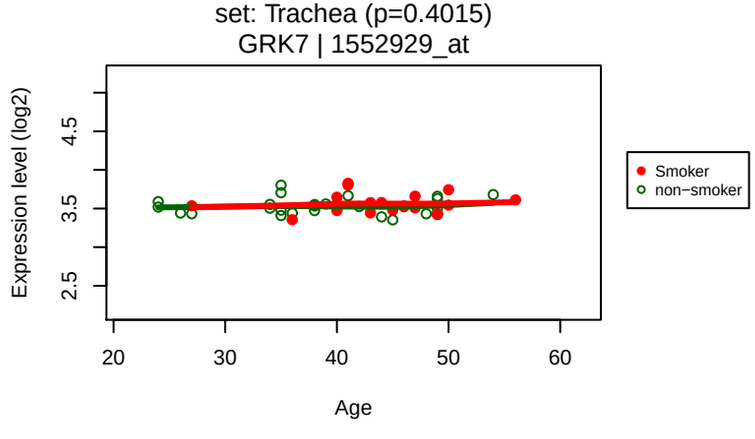

Gene expression ratio in Males
gene: GRK7 | 1552929_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5092
p=0.9968
p=0.4943
p=0.77
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRK7 | 1552929_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5479
p=0.0318
p=0.8203
p=0.2242
SmallAirway2
LargeAirway2
Trachea
Macrophage2

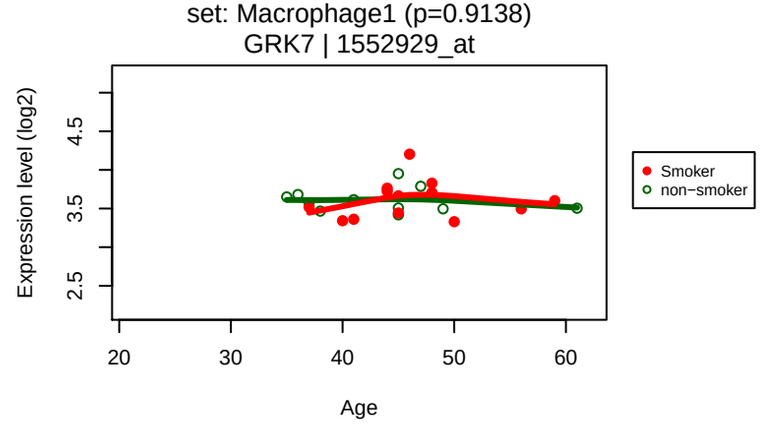

set: SmallAirway1 (p=0.9648)
GRK7 | 1552929_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker

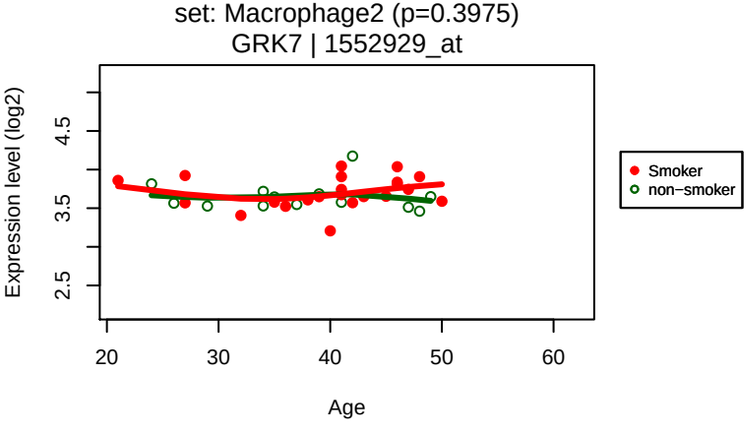

set: SmallAirway2 (p=0.9337)
GRK7 | 1552929_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker

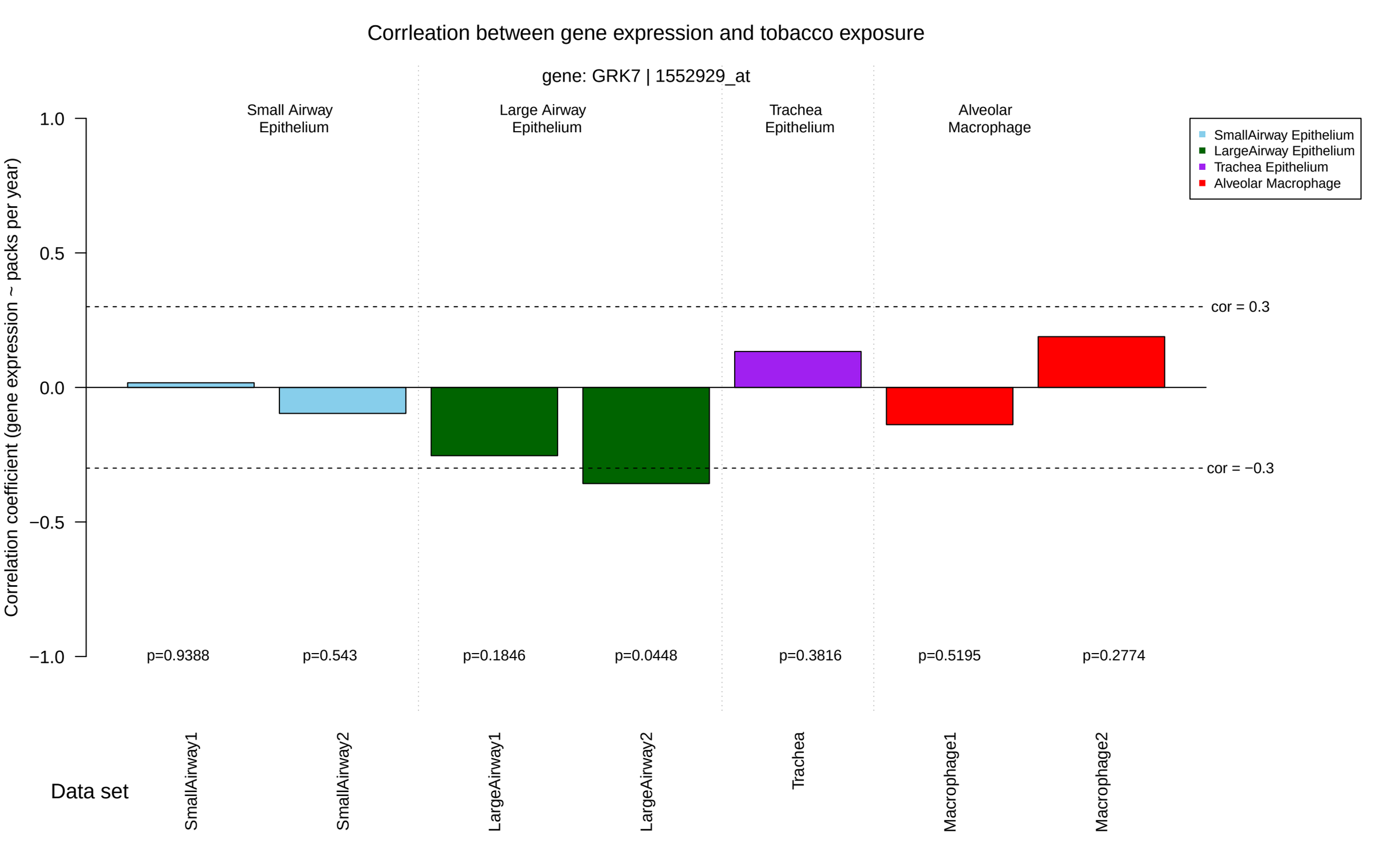

set: LargeAirway1 (p=0.3207)
GRK7 | 1552929_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3142)
GRK7 | 1552929_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.4015)
GRK7 | 1552929_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9138)
GRK7 | 1552929_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3975)
GRK7 | 1552929_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRK7 | 1552929_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9388
p=0.543
p=0.1846
p=0.0448
p=0.3816
p=0.5195
p=0.2774
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2